LABORATORY TITLE: IOS FORENSICS

STUDENT NAME: WALIYAT OLALEKAN

STUDENT ID: 2025/FWSD/11435

COURSE NAME: MOBILE AND IOT FORENSICS CASE STUDY

COURSE CODE: SBT-DF205 (LAB 2)

INSTRUCTOR NAME: MR. AMINU IDRIS

DATE OF SUBMISSION: NOVEMBER 5TH, 2026.

VERSION: Version 1.0

Executive summary

The examination identified device and account configuration artefacts, incoming SMS messages, contact records, call-history entries, Safari browsing history, calendar data, photo-library metadata, KnowledgeC activity records, Health source-device metadata and routine learned-location records. The principal date range represented by the examined artefacts is 21 March to 16 April 2020, with Safari activity specifically observed on 28 March 2020.

The selected call-history entries have zero duration and are marked unanswered/not originated; they do not establish completed conversations. The Calendar event “Pick up lunch” is a scheduled entry, not proof that the event occurred. Health database source metadata identifies iPhone and Apple Watch devices, but no workout sessions or workout statistics were found. Several expected artefacts were not located in the selective extraction.

Overall, the data supports a multi-source activity timeline, but several interpretations remain limited by missing media files, unavailable encrypted location/Health databases, undecoded numeric codes, and the absence of a pre-extraction archive hash.

Scope and examination

The examination addressed the assignment's device/account, communications, browsing/calendar, photo/media, KnowledgeC, Health/location and timeline requirements. The work was limited to artefacts and command outputs actually examined in the supplied selective extraction. It did not attempt to attribute every action to a specific human, recover unavailable files, decode undocumented numeric enums, or infer intent from app identifiers.

The supplied assignment identifies the source as an iPhone SE iOS 13.4.1 case. Device identifiers and the observed artefact set are reported separately from conclusions about the person using the device.

Task 1 Evidence preservation

The original ZIP and TAR paths were recorded in the case workspace. SHA-256 values supplied from the examination are:

ios 13_14.1.zip - 9502ecbea02cc890b11f1d95a9f9a1e2287b2ddcc8be5257d56cc57aacee71c2;

ios 13_14.1.tar - 991727d92145a08623e998c008cc2a05f086be722ee2945815d0eb46a78f04fb.

An earlier full extraction was incomplete: it contained 111,459 files, while its private directory was effectively empty apart from .DS_Store files; the var symlink consequently pointed to an unavailable private/var path. That extraction was not used for user-data conclusions. A later selective extraction was created under the correct case workspace at SBT-DF205-C2/Working/Selective_Extraction/Volumes/JOSH/NoTar-13-4-1. Selected SQLite databases and available -wal/-shm sidecars were retained together.

During TAR listing/extraction, the tool reported “Ignoring unknown extended header keyword 'SCHILY.fflags'.” This was recorded as an extended-header warning, not treated by itself as proof of archive corruption. A post-extraction hash chronology and the incomplete earlier extraction are material limitations and must remain disclosed.

SQLite examination used schema inspection, record counts, selected-row queries and integrity/quick checks were recorded. Reported successful checks include Accounts3, CallHistory, Safari, Calendar, Photos, KnowledgeC, Health and Local databases.

Task 2 Device accounts and settings

SystemVersion.plist records ProductVersion 13.4.1 and ProductBuildVersion 17E262. Health source_devices records an iPhone with product identifier iPhone8,4 and software 13.4.1; this identifier is consistent with the first-generation iPhone SE. The same source-device table also contains an iPhone8,4 record with software 13.3.1, and Apple Watch Watch4,3 records with software versions 6.1.3 and 6.2.1. These are source-device records and should not be conflated with the current system version plist. The global preferences record Apple language en-US and locale en_US. SpringBoard preferences record battery percentage display enabled and ringer muted set to false. These are stored configuration values, not proof that the device was used in a particular manner.

Accounts3.sqlite passed quick_check and contained 18 account records. Service labels included iTunes Store, Messages, iCloud, Find My Friends, Device Locator, CloudKit, IMAPNotes, CalDAV, CardDAV, IDMS, Apple ID, Game Center and Gmail. Some records shared masked account identifiers or represented service-level entries. The count must not be interpreted as 18 distinct people.

Task 3 Messages contacts and calls

The examined sms.db query returned ten incoming SMS records (IDs 1–10) dated 22–25 March 2020. The sender/service labels included Google, imo, LINE, MeWe, Signal, Facebook, Telegram, TikTok, WhatsApp and Skout. Several messages were verification or account-notification related. This supports the conclusion that the device received such messages; it does not prove that the associated accounts were successfully accessed, that a code was used, or that the device owner initiated the related activity.

AddressBook.sqlitedb yielded two contact records. The retrieved display labels were “Josh” and “This Is DFIR”. One record had two phone-value rows and one email-value row; the values are redacted. The second record had no phone or email value in the retrieved output

CallHistory.storedata passed its integrity check and contained 37 ZCALLRECORD rows. Three selected records were examined: IDs 12, 16 and 26, dated 23, 24 and 26 March 2020 respectively. Each had a recorded duration of 0.0 seconds, answered=0 and originated=0. The records also contained calltype=1, but the enum meaning was not decoded. Accordingly, they are reported as call-history entries with unanswered/not-originated flags, not as completed conversations.

The evidence establishes receipt of multiple SMS messages, the existence of two contact records and the presence of call-history entries. It does not establish successful account authentication.

Task 4 Safari and calendar

Safari History.db passed its integrity check and contained ten history_items and sixteen history_visits. Ten visit rows were examined. On 28 March 2020, history records include a Google search concerning when the 2020 MLB season would start, an MLB.com page concerning the season delay, a Google search asking whether the NHL would resume, and an NHL.com article about resumption after the coronavirus pause. Several items were visited more than once. A history record demonstrates a recorded browser visit, not the identity of the person at the device or their reason for visiting.

Calendar.sqlitedb passed its integrity check and contained 119 CalendarItem rows. The examined grouping showed 118 entries associated with US Holidays and one home calendar event. Calendar item ID 119 was titled “Pick up lunch” and scheduled for 12 April 2020, 14:00–15:00 UTC, with America/New_York stored as its time zone. The record is evidence of a calendar entry, not proof that the lunch was collected or that the device visited a corresponding place. CalendarExtras was not located in the selective extraction

Task 5 Photographs and media

The ZGENERICASSET table contained 59 asset records (58 with ZKIND=0 and one with ZKIND=1). The first ten examined records included IMG_0001.JPG through IMG_0010.JPG, with dimensions, filenames, directory references, capture/addition timestamps and related metadata. Three examples are:

IMG_0001.JPG recorded at 22 March 2020 13:05:24.541041 UTC (1125×1137); IMG_0003.JPG at 13:10:50.327283 UTC (1668×860); and IMG_0009.JPG at 13:31:03.109 UTC (828×637), with an addition timestamp of 13:47:01.964396 UTC.

The first ten asset records contained latitude/longitude values of (-180,-180). These were treated as unavailable sentinel values, not valid GPS coordinates. IMG_0009 and IMG_0010 contained EXIF timestamp strings consistent with their recorded capture times after applying the recorded -04:00 offset. However, the corresponding media files were not present in the selective media extraction. No ExifTool validation, visual inspection or media-file hashing could therefore be completed.

Task 6 KnowledgeC activity

The /app/usage records provide start/end intervals and app identifiers. Examples on 16 April 2020 include MobileSMS (15:57:15–15:59:13 UTC), Picaboo (15:59:16–16:00:09), Cyberdust (16:00:10–16:01:00) and Photos (16:02:52–16:03:26). The broader query summary also showed repeated intervals for Preferences, Mail, Messages, App Store, Photos, FitbitMobile, Home, Phone, Burner, Podcasts, Fitness and Zoom.

App-use intervals are useful timeline indicators but do not establish message content, what was viewed, the user's intent, or exact screen time. The /activity/level and /inferred/motion streams contained numeric values, but their quantity-type and metadata fields were null in the examined

rows and the value-code semantics were not decoded. Those values are therefore preserved as raw numeric codes and are not labelled as walking, running, exercise or any other specific activity.

Task 7 Health and location evidence

healthdb.sqlite contained six source records and eight source-device records. Device records included iPhone8,4 and Apple Watch Watch4,3, as well as AirPods. The database contained zero workout_sessions, zero workout_builder_statistics and zero workout_session_events; related workout-builder tables were also empty. The examined database therefore did not provide two verified Health activity samples or workout measurements. healthdb_secure.sqlite was not located in the selective extraction. The existence of a source-device record does not prove that a wearable was worn during a particular event or that activity was synced at a particular time.

Local.sqlite contained five ZRTLEARNEDLOCATIONOFINTERESTMO records, fifteen ZRTLEARNEDLOCATIONOFINTERESTVISITMO records and fifteen ZRTLEARNEDLOCATIONOFINTERESTTRANSITIONMO records. The visit intervals span 24 March to 16 April 2020. For example, Visit ID 971 (Location ID 271) begins 25 March 2020 16:41:25.011714 UTC and ends 27 March 2020 18:57:46.726572 UTC. Location ID 271 contained 130,092 datapoints and a confidence value of 1.0, with a record-creation time of 16 April 2020 16:55:14.962794 UTC. The learned-location record creation timestamp is not the time of the underlying visit. The encrypted location cache_encryptedB.db was not located in the selective extraction. The transition query displayed four rows before stopping because one timestamp could not be converted (year out of range).

Task 8 Timeline and conclusions

UTC	Sources/ID	Observation
2020-03-22 01:10:59 UTC	SMS / Message ID 1	incoming SMS received; Google verification message. Message content/code intentionally redacted.
2020-03-22 13:05:24.541041 UTC	Photos / Asset IMG_0001.JPG	Photo-library metadata records capture time; 1125x1137; no usable GPS coordinates.
2020-03-22 13:10:50.327283 UTC	Photos / Asset IMG_0003.JPG	Photo-library metadata records capture time; 1668x860.
2020-03-22 13:47:01.964396 UTC	Photos / Asset IMG_0009.JPG	Asset added to library; EXIF local time 09:31:03 with UTC offset -04:00, consistent with recorded UTC capture time 13:31:03Z.
2020-03-23 20:02:52.413675 UTC	calls / Call record ID 12	Call record with zero duration; answered=0 and originated=0.
2020-03-24 17:37:18.386421 UTC	calls / Call record ID 16	Call record with zero duration; answered=0 and originated=0.
2020-03-25 16:41:25.011714 UTC	Location / Visit ID 971 / Location ID 271	Learned-location visit interval begins; recorded exit 2020-03-27T18:57:46.726572Z
2020-03-26 17:51:45.317984 UTC	calls / Call record ID 26	Call record with zero duration; answered=0 and originated=0.

2020-03-28 00:58:36.117145 UTC	Safari / Visit ID 2 / History item ID 5	Safari history records Google search concerning when the 2020 MLB season would start
2020-03-28 00:59:03.212165 UTC	Safari / Visit ID 3 / History item ID 5	Safari history records MLB.com page titled about the 2020 season being delayed.
2020-03-28 01:02:44.282689 UTC	Safari / Visit ID 9 / History item ID 5	Safari history records Google search asking whether the NHL would resume.
2020-03-28 01:03:51.330998 UTC	Safari / Visit ID 11 / History item ID 5	Safari history records NHL.com article concerning resumption after the coronavirus pause.
2020-04-01 17:57:57.994081 UTC	Location / Visit ID 958 / Location ID 272	Learned-location visit interval begins; exit 18:00:55.994825Z
2020-04-04 22:04:59.997631 UTC	Location / Visit ID 962 / Location ID 272	Learned-location visit interval begins; exit 22:09:32.994294Z.

Conclusion

The supplied extraction contains multiple independent artefact families that support a dated activity chronology from March to April 2020. The system plist records iOS 13.4.1 build 17E262, and Health source-device metadata includes iPhone8,4, consistent with first-generation iPhone SE. The examined SMS database contains incoming verification-related messages; Safari history records MLB/NHL-related browsing; Calendar contains one home event; Photos.sqlite contains 59 asset metadata records; KnowledgeC contains extensive app and device-state records; and Local.sqlite contains learned-location visits.

The evidence does not establish successful use of accounts mentioned in verification messages, completed conversations for the selected call records, the content of unavailable photographs, specific physical activities from undecoded numeric codes, actual completion of the calendar event, or definitive human presence at learned locations. Health workout/sample records and two expected encrypted databases were not available in the selective extraction. These limitations materially constrain attribution and interpretation.